

INFORMATION VISUALISATION — EXAM CHEATSHEET

COMPSCI5099 · University of Glasgow · 07 May 2026 · 90 min · 4 × 15 marks · Closed Book

DATA TYPES

Attribute Type	Operations	Example
Nominal	= ≠	Country, Ward, Blood type
Ordinal	= ≠ < >	Low/Med/High, Grades
Quantitative	= ≠ < > + −	Age, Weight, Salary

7 Data Types (1-2-3-T-M-Tr-N): 1D · 2D · 3D · Temporal · Multi-dim · Tree · Network

BERTIN'S VISUAL VARIABLES

Variable	Nominal	Ordinal	Quant	Select?
Position	✓	✓	✓ BEST	✓
Size	■	✓	✓	✓
Shape	✓	✗	✗	✗ ONLY
Hue	✓	✗	✗	✓
Value (light)	✗	✓	■	✓
Orientation	■	✗	✗	✓
Texture	✓	■	✗	✓

★ Shape = only non-selective. Position & Size = only quantitative. Hue → nominal. Value → ordinal.

SHNEIDERMAN'S MANTRA + 7 TASKS

"Overview first, zoom and filter, then details-on-demand"

Task	When to use
① Overview	Always first — show all data
② Zoom	User spots something interesting
③ Filter	Remove irrelevant items
④ Details-on-demand	Click/hover for full record
⑤ Relate	Show connections between items
⑥ History	Undo / replay exploration
⑦ Extract	Export filtered subset (e.g. CSV)

STEVENS' LAW + CHANNEL ACCURACY RANKING

$P = k \times S^n$ ($n \approx 1.0$ = accurate, $n < 1.0$ = underestimated)

Rank	Channel	n	Example chart
① BEST	Position (common scale)	≈ 1.0	Bar chart, Scatter plot
②	Position (unaligned)	≈ 1.0	Grouped bar chart
③	Length / Height	≈ 1.0	Bar (the bar itself)
④	Angle / Slope	—	Pie chart ← WHY PIES FAIL
⑤	Area	≈ 0.7	Bubble (30% underestimated)
⑥	Volume	≈ 0.6	3D charts ← always avoid
⑦ WORST	Colour saturation	≈ 0.5	Heatmap

COLOUR THEORY

Map Type	Use for	Example
Sequential	Ordered, one direction	Light→dark blue for rainfall
Diverging	Meaningful zero/midpoint	Blue→white→red for anomaly
Categorical	Nominal (≤ 8 categories)	Countries, product types

Rainbow = BAD — not perceptually uniform, false boundaries. Use Viridis / Inferno. Never red+green only (8% males colour-blind). Use redundant encoding (colour + shape).

PRE-ATTENTIVE + CHART TYPES

<200ms, no attention needed = **pop-out**. Single feature (colour/size/orientation) → parallel. Two features combined → serial scan (NOT pre-attentive). Gestalt: proximity, similarity, continuity, enclosure.

Chart	Use for — Avoid when
Bar	Compare categories — not trends over time
Line	Trends over time — not categorical x-axis
Scatter	Correlation/clusters — not categorical data
Pie	Rough proportion ≤ 5 — never precise comparison
Histogram	Distribution shape — not comparing groups
Box plot	Distribution summary — hides bimodality
Heatmap	Large matrix patterns — not precise values

MULTIDIMENSIONAL + DIM REDUCTION

Technique	Dims	Strength	Weakness
Parallel Coords	10–20	All dims + correlations	Adjacent axes only
SPLOM	8–12	All pairwise plots	$n \times (n-1)$ plots, small
Heatmap	50+	Compact, large data	Colour limits precision
t-SNE / UMAP	100+	High-D overview	Loses dim meaning

Parallel coords: parallel lines=+ve, X-shape=−ve correlation. **SPLOM:** 9 dims = 72 plots. **Dim reduction:** PCA=linear, t-SNE=local only, UMAP=local+global (best).

NETWORK + TEMPORAL

	Node-Link	Matrix
Best for	Sparse, path tracing	Dense, connection check
"A→B connected?"	Hard — visual search	Easy — row A, col B
Weakness	Hairball when dense	Can't trace paths

Temporal Technique	Use when
Line chart	Single series, trends
Small multiples	Compare across categories
Calendar heatmap	Cyclic / day-of-week patterns
Gantt chart	Intervals and durations

Animation vs Small Multiples: Animation = change blindness (relies on memory). Small multiples = direct comparison. Always use small multiples for analysis tasks.

LARKIN & SIMON (1987)			EVALUATION — MUNZNER'S NESTED MODEL		
Concept	Meaning		Level	Question asked	Threat if wrong
Informational Equiv.	Same information in both representations		1. Domain	What do users need?	Solving wrong problem
Computational Equiv.	Same cognitive effort to use both		2. Abstraction	Right data types & tasks?	Wrong data model
3 Diagram Advantages	Meaning		3. Encoding	Right marks & channels?	Ineffective visualisation
Locality	Related info grouped → less search needed		4. Algorithm	Efficient implementation?	System too slow to use
Minimise labelling	Spatial position carries meaning implicitly		Eval Method		Use when
Perceptual inference	Visual system does inference (e.g. lines cross)		Controlled experiment		Prove A faster/better than B statistically
★ Key: Two reps can be informationally equivalent but NOT computationally equivalent. Vis amplifies cognition by reducing cognitive effort — not by adding information.			Think-aloud / interview		Understand WHY users struggle
			Heuristic evaluation		Quick expert design review
			Case study / deployment		Real users, highest ecological validity
HEER — GENERALISED SELECTION			TUFTE + PITFALLS		
	Dynamic Queries	Query Relaxation (Heer)	Tufte Rule		Meaning
Direction	Broad → narrow	Specific → broad (bottom-up)	Data-ink ratio		Maximise — remove all non-data ink
Method	Sliders / toggles	Click example → generalise	Chartjunk		Delete 3D effects, heavy grids, decoration
Errors?	Possible	Structurally impossible	Lie factor		Graphic effect ÷ data effect ≈ 1.0
			Small multiples		Better than animation for comparison
			✗ Pitfall		✓ Fix
			Truncated y-axis		Start bar charts at 0 always
			Pie for comparison		Use bar chart (position beats angle)
			Rainbow colour map		Use Viridis / Inferno / Magma
			Hue for ordinal data		Use colour value (lightness)
			Unjustified 3D		Use 2D — 3D = occlusion + distortion

KEY PEOPLE & PAPERS			
Person	Year	Key Contribution	★ Exam Hook
Bertin	1967	Visual variables: position + 6 retinal. Associative/selective/ordered/quantitative.	Shape=only non-selective. Position=most powerful.
Shneiderman	1996	"The Eyes Have It" — Mantra + 7 tasks × 7 data types.	Overview first, zoom and filter, details-on-demand
Tufte	1983+	Data-ink ratio, chartjunk, lie factor, small multiples.	Above all else, show the data
Munzner	2015	Nested model (4 levels). Marks & channels. Effectiveness ranking.	Expressiveness + Effectiveness
Cleveland & McGill	1984	Perceptual accuracy hierarchy: position > length > angle > area > volume.	Position on common scale = most accurate
Larkin & Simon	1987	"Why a Diagram is (Sometimes) Worth 10,000 Words" — info vs computational equiv.	Same info, different cognitive effort
Heer et al.	~2008	Generalised selection via query relaxation. Selection = declarative query.	Bottom-up, example-driven query building
Treisman	1985	Feature Integration Theory. Single feature = parallel. Conjunction = serial scan.	Single features pop out; combinations do not
Stevens	1957	Power Law P=kS^n. Area (n=0.7) underestimated. Position (n=1.0) accurate.	Not all channels encode quantity equally

WHEN TO APPLY WHAT — Quick Exam Decision Guide		
IF the scenario involves...	APPLY...	Person
Choosing which visual property (colour/size/shape) to encode data	Bertin visual variables — match variable to data type	Bertin
Why one chart is more/less accurate than another	Stevens law + Cleveland & McGill ranking — state n and rank number	Stevens + C&M;
Designing interaction for a complex system	Shneiderman mantra + 7 tasks — overview → filter → details-on-demand	Shneiderman
Why a diagram / vis helps more than text or prose	Larkin & Simon — informational vs computational equiv, 3 advantages	Larkin & Simon
How users select and query data interactively	Heer generalised selection vs dynamic queries — declarative query + relaxation	Heer
Data has 5+ dimensions / many attributes	Parallel coordinates or SPLOM — brushing, axis reordering, correlations	Munzner
Visualising a network, graph, or hierarchy	Node-link vs matrix — choose based on graph density + task type	Graph theory
Data changes over time	Line chart / small multiples. Animation = change blindness. Small multiples win.	Tufte
Evaluating whether a vis system works	Munzner nested model (4 levels + threats). Quant (measure) vs qual (understand why).	Munzner
Critiquing a bad visualisation	Tufte: data-ink ratio, chartjunk, lie factor. Check pitfalls: rainbow, 3D, truncated axis.	Tufte

EXAM STRATEGY	Every 15-mark answer: ① Classify data type → ② Choose marks & channels, justify with accuracy ranking → ③ Apply Shneiderman mantra to interaction → ④ Name the person / paper → ⑤ State pitfalls. Prof's advice: Think "How would I USE this here?" not "What are all the details?"
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